

# Strategic Under-Specification: A Stackelberg Game Between User and Assistant

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## Abstract

Users of coding assistants routinely under-specify their intent, and the assistant’s policy for handling missing information (whether guessing silently or asking a clarifying question) is an endogenous strategic choice that users respond to. We develop a contract-theoretic model of endogenous clarification in human–AI interaction: strategic under-specification is not driven by models becoming better at guessing; it emerges because users strategically economize on specification effort in response to the assistant’s endogenous asking policy when the assistant’s objective diverges from true user welfare. Formalizing this interaction as a Stackelberg game between an assistant (leader) and a population of users with privately known specification costs (followers), we first establish a decision-theoretic benchmark with an exogenous policy. Here, improving raw guessing accuracy alone cannot harm users despite inducing them to specify less. This result provides an immediate envelope-theorem benchmark, showing that informal claims that better guessing degrades outcomes conflate technological improvement with policy misalignment.

Closing the loop, we solve for the assistant’s optimal policy under two governance regimes: a *pooling* policy restricted to a single population-wide ask rate, and a *screening* policy offering an incentive-compatible menu conditioned on prompt specification effort. We show that an unbiased (welfare-aligned) assistant achieves full first-best efficiency under screening despite users’ private information, as incentive compatibility is costless when principal and agent share an objective. Consequently, all screening-regime inefficiency is attributable to misalignment between the assistant’s objective and true user welfare, not to asymmetric information per se. Pooling, by contrast, suffers a structural loss from restricted instruments even when perfectly aligned, collapsing to an extreme bang-bang corner solution. We characterize both equilibria under a tractable linear-risk specification, derive testable comparative statics, verify that the qualitative predictions survive under the exact nonlinear conjunctive specification, establish a five-level hierarchy of theoretical and computational evidence, and outline a proposed human-subject experimental design.

## 1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user’s specification effort depends on what they expect the assistant to do with silence, and the assistant’s ask/guess policy is chosen (explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Its central premise is that strategic under-specification is not caused by AI getting better at guessing; it

is caused by users responding to the assistant’s endogenous clarification policy when the assistant’s objective differs from theirs. Section 2 begins with a decision-theoretic benchmark where the assistant’s policy is exogenous, proving that improved guessing accuracy alone weakly improves user welfare despite shortening prompts. Section 3 identifies the true moral-hazard channel: unobserved divergence between the user’s belief and the assistant’s true asking policy. Sections 4–7 close the loop by modeling the assistant as a Stackelberg leader anticipating user best responses under pooling and screening instruments. Section 8 translates the equilibrium characterizations into seven falsifiable behavioral predictions (P1–P7). Section 10 evaluates exact nonlinear robustness and discusses institutional extensions.

## 1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

**Contract theory and screening without transfers.** Our screening framework builds on classical adverse-selection models (Myerson, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate  $a$  offered in response to specification effort  $m$ ). This connects to non-monetary screening and costly verification (Townsend, 1979), where information revelation is governed by operational friction rather than cash transfers.

**Costly communication and strategic disclosure.** The user’s specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user’s rational economizing on specification costs when anticipating automated assistance.

**Assistance games and human–AI interaction.** Our bi-level interaction connects to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016; von Stackelberg, 1934), where an autonomous assistant acts to maximize human utility under incomplete information. However, while assistance games formalize the assistant’s uncertainty over the human’s underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value  $V$  and error loss  $L$ ). Instead, the informational asymmetry resides entirely in the user’s private *specification cost*  $\kappa$ . The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. To our knowledge, our model provides the first contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

**Clarification and interactive NLP.** A parallel empirical literature in natural language processing and information retrieval investigates algorithms for generating and ranking clarifying questions in conversational search and dialog (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020). These works optimize static clarification decisions or train heuristic dialog policies via reinforcement learning against fixed datasets. However, they uniformly treat user inputs as exogenous test instances rather than the output of an optimizing agent who strategically anticipates

the clarification policy. Our framework provides the contract-theoretic microfoundation that this engineering literature lacks, formally endogenizing prompt specification effort as an equilibrium best response to the assistant’s asking policy.

## 2 Environment and the Decision-Theoretic Benchmark

### 2.1 Setup

A task has  $k$  binary, independent, intent-relevant attributes  $\theta = (\theta_1, \dots, \theta_k) \in \{0, 1\}^k$ , each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level  $m \in \{0, \dots, k\}$  and truthfully reveals  $m$  attributes.<sup>1</sup> Specifying  $m$  attributes costs  $c(m)$ , with  $c(0) = 0$ ,  $c' \geq 0$ ,  $c'' \geq 0$ .

For each of the  $k - m$  unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability  $g \in (0, 1)$ . We model the assistant’s policy at this stage as a uniform ask rate  $a \in [0, 1]$  applied to every unspecified attribute, so each is correctly resolved with probability  $q(a, g) = a + (1 - a)g$ . Task success requires *all*  $k$  attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user’s payoff is

$$U(m; a, g) = V q(a, g)^{k-m} - c(m) - a(k - m)\kappa,$$

with  $V > 0$  the value of success and  $\kappa > 0$  the user’s cost of answering one clarifying question. We treat  $m$  as continuous on  $[0, k]$  for tractability.

### 2.2 Proposition 1: accuracy alone does not create moral hazard

**Proposition 1.** *Fix  $a$ . Let  $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$ . Then  $U^*$  is weakly increasing in  $g$ .*

*Proof.* By the envelope theorem, at the optimum  $m^*(g)$ ,  $dU^*/dg = \partial U / \partial g|_{m=m^*(g)} = V(k - m^*)q^{k-m^*-1}(1 - a) \geq 0$ . The envelope theorem lets us ignore  $dm^*/dg$ : because  $m^*$  is chosen optimally, its first-order response to  $g$  has no first-order effect on  $U^*$ .  $\square$

**Assumption 1** (Regularity). *In the relevant range of  $m$ ,  $(k - m) \ln q(a, g) \geq -1$ .*

**Remark 1** (Interpreting Assumption 1, and Monotonicity Survival). *The condition has no closed-form economic reading on its own; it is a sufficient condition for the single-crossing property used in Corollary 1 and Proposition 2 via Topkis’s monotone comparative statics theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994), not a claim about primitives directly. The direction is pinned down by the cross-partial itself: for fixed  $a$ ,*

$$\frac{\partial^2 U}{\partial m \partial g} = -V(1 - a)q^{k-m-1}[(k - m) \ln q + 1],$$

*so  $U$  has decreasing differences in  $(m, g)$ , which is the condition required by Topkis’s theorem (Topkis, 1978, 1998) to ensure that  $m^*(g)$  is weakly decreasing, exactly when  $(k - m) \ln q + 1 \geq 0$ , namely when Assumption 1 holds as stated here. If the inequality were reversed,  $U$  would exhibit increasing differences and hence  $m^*(g)$  would be locally increasing (e.g., at  $V=10, k=5, a=0, c(m)=m^2/2$ ,  $m^*$*

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<sup>1</sup>We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

risers from  $\approx 0.26$  at  $g=0.5$  to  $\approx 0.67$  at  $g=0.9$  before turning over and falling as  $g \rightarrow 1$ ). Loosely, since  $\ln q \approx -(1-q)$  for  $q$  near 1, the condition is close to  $(k-m)(1-q) \leq 1$ : the expected number of wrong attributes under the current policy is at most about one. A numerical grid evaluation over  $k \in [5, 15]$ ,  $q \in [0.70, 0.95]$ , and  $\kappa \in [0.20, 2.00]$  (726 grid points, quadratic cost  $c(m) = \kappa m^2/2$ , evaluated at each point's own optimizing  $m^*$ ) finds that under the benchmark simulation valuation ( $V = 100$ ), the condition holds on 89.9% (653/726) of this parameter space. In lower-valuation regimes where specification effort is lower ( $V = 10$ , as in the numerical example above), the condition holds on roughly 34% (and 31.7% at  $V = 8$ ); the exact share is sensitive to the scaling of  $c(m)$  relative to  $V$ . Across the tested parameter grid (2,552 parameter intervals), a comprehensive finite-difference evaluation confirms that monotonicity of specification ( $m^*(g)$  weakly decreasing in  $g$ ) holds without exception across all 2,162 intervals where Assumption 1 is satisfied (100% survival, zero violations). Outside the sufficient condition ( $(k-m) \ln q < -1$ , 390 intervals), the cross-partial turns strictly positive and monotonicity breaks in 96.7% of tested intervals (377/390), confirming that the regularity condition is empirically tight.

**Corollary 1.** *Under Assumption 1,  $m^*(g)$  is weakly decreasing in  $g$  (decreasing differences in  $(m, g)$ ; Topkis's theorem).*

*Proof.* Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition  $\partial^2 U / \partial m \partial g \leq 0$  on the relevant range, which is decreasing differences in  $(m, g)$ ; Topkis's theorem then gives  $m^*(g)$  weakly decreasing.  $\square$

Thus, holding the assistant's policy fixed, users specify less as the assistant becomes more adept at guessing, yet they are never worse off for doing so. Proposition 1 serves primarily as a baseline: it applies the envelope theorem to a single agent optimizing against a technology parameter that enters the payoff with a non-negative derivative, a property that holds generally in models of this class. Its purpose is not to present a surprising result, but to establish the rigorous benchmark against which claimed moral hazard must be evaluated, ensuring that Proposition 2 isolates the true source of welfare loss. This frames the central question of this paper: what happens once the policy  $a$  is no longer exogenous, but chosen by a strategic assistant that anticipates the user's response?

### 3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant's ask rate is fixed and that users correctly anticipate  $q(a, g)$ . Both assumptions can fail.

#### 3.1 Cost externality

If a third party (such as a code reviewer or downstream production system) bears part of the cost of an erroneous guess, the user's private parameters  $(\kappa, V)$  understate the social costs, causing the user to under-specify relative to the social optimum at any fixed  $g$ , an inefficiency that worsens as  $g$  rises by Corollary 1. While this constitutes a standard Pigouvian externality, our primary focus lies in strategic and informational friction.

#### 3.2 Miscalibrated belief updating under silent failure

If incorrect guesses are not always detected, and the detection probability  $d(m)$  is increasing in  $m$ , users infer  $g$  from experience rather than observing it directly. Under-specifying users are

consequently less likely to catch their own errors, generating an upward-biased estimate  $\hat{g}$  that reinforces under-specification. This mechanism represents a dynamic learning process rather than a static equilibrium outcome; we formalize its testable implications in Section 8 (prediction P5).

### 3.3 Belief about the assistant’s policy

Let users be *sophisticated* (correctly infer the assistant’s true ask rate  $a_A$ ) or *naive* (believe the assistant follows the welfare-optimal rate  $a^\dagger \geq a_A$ ).

**Proposition 2.** *Under Assumption 1, naive users choose  $m_{naive}^* \leq m_{soph}^*$  and  $U(m_{naive}^*; a_A, g) < U(m_{soph}^*; a_A, g)$ : naive users under-specify and are strictly worse off than sophisticated users, holding raw guessing accuracy  $g$  fixed throughout.*

*Proof.* Because  $q(a, g) = a + (1 - a)g$  is strictly increasing in  $a$  for  $g \in (0, 1)$ , a naive user believing  $a^\dagger \geq a_A$  perceives a higher resolution accuracy  $q(a^\dagger, g) \geq q(a_A, g)$ . Under Assumption 1, Corollary 1 proves that  $m^*(g)$  is weakly decreasing in accuracy, so  $m_{naive}^* \leq m_{soph}^*$ . Furthermore,  $U(m; a_A, g)$  is strictly concave in  $m$  with unique maximizer  $m_{soph}^*$ , so any choice  $m_{naive}^* \neq m_{soph}^*$  delivers strictly lower true utility:  $U(m_{naive}^*; a_A, g) < U(m_{soph}^*; a_A, g)$ .  $\square$

This identifies the first true source of moral hazard: not technological accuracy, but an unobserved gap between the user’s belief regarding the assistant’s policy and the assistant’s actual policy. Section 5 solves for the optimal policy  $a_A$  when the assistant acts as a strategic, potentially biased Stackelberg leader, allowing Proposition 2 to be evaluated with an endogenously derived policy rather than an exogenously assumed one.

## 4 The Assistant’s Policy as a Stackelberg Game

### 4.1 General formulation

The assistant (leader) commits ex ante to a policy  $\pi$  before meeting any user. A population of users (followers) has privately known specification-cost type  $\kappa \sim F$  on  $[\kappa_L, \kappa_H]$ , with cost  $c(m; \kappa) = \kappa m^2/2$ , and best-responds  $m^*(\kappa; \pi) = \arg \max_m U(m; \pi, \kappa)$ . The leader’s own objective  $\Pi$  need not equal  $U$ : we write

$$\Pi_\kappa(m, a) = \mu_A U(m, a; \kappa) - (\lambda_A - c_Q) a (k - m),$$

where  $\mu_A \in (0, 1]$  discounts the leader’s weight on true downstream user welfare,  $c_Q$  is the user’s true cognitive cost of answering one clarifying question, and  $\lambda_A$  is the leader’s own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ( $\lambda_A > c_Q$ ): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (e.g., DPO) commonly evaluate conversation turns in isolation. Reward models systematically penalize “evasive” clarifying questions in favor of immediate code generation because users in rating pools favor direct answers. Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side pressure to suppress clarification turns.
2. *Downstream welfare discounting* ( $\mu_A < 1$ ): Turn-level user satisfaction signals (such as prompt acceptance or immediate positive feedback) capture immediate session fluency. In contrast, the true economic loss  $L$  from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, test suites, or code review.

Consequently, training objectives tuned to immediate session retention tend to discount downstream correctness.

The leader is *unbiased* iff  $\mu_A = 1$  and  $\lambda_A = c_Q$ , in which case  $\Pi_\kappa \equiv U(\cdot; \kappa)$  exactly. The leader solves  $\pi^* = \arg \max_\pi \mathbb{E}_{\kappa \sim F}[\Pi_\kappa(m^*(\kappa; \pi), \pi)]$ , a bilevel program.

**Remark 2** (Existence). *For Model I below ( $\pi = a \in [0, 1]$ , a compact scalar policy),  $m^*(\kappa; a)$  is continuous in  $a$  (strict concavity of  $U$  in  $m$  plus the implicit function theorem), so  $\mathbb{E}_\kappa[\Pi_\kappa]$  is continuous in  $a$  and a maximizer exists on  $[0, 1]$  by Weierstrass's theorem. For Model II below (a finite menu over two types), the feasible set is a compact subset of  $\mathbb{R}^4$  cut out by linear IC/IR constraints and a continuous objective, so a maximizer again exists by Weierstrass's theorem; no ironing or ex-ante randomization is needed with only two types.*

For tractability in what follows we replace the exact conjunctive success probability  $q^{k-m}$  with its first-order (small-risk) approximation around  $q \rightarrow 1$ : expected loss  $\approx L(1 - q)(k - m)$ , where  $L$  denotes the economic loss from a single unresolved-and-incorrect attribute. With  $\Lambda \equiv L(1 - g)$ , the user's payoff becomes

$$U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - a c_Q(k - m).$$

Throughout Sections 4.2–6, all closed-form characterizations are stated for the interior domain  $\kappa \geq \Lambda/k$ , ensuring that the unconstrained optimal specification  $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa}$  lies strictly in  $[0, k]$  for all  $a \in [0, 1]$  under the maintained condition  $\Lambda > c_Q$ . For low-cost types ( $\kappa < \Lambda/k$ ), best-response effort saturates at the physical attribute capacity  $m^* = k$ , resolving all attributes upstream. As demonstrated in Section 10 and verified in our computational test suite, the qualitative regime structure and corner-solution properties survive boundary clipping identically.

## 4.2 The first-best benchmark

Before analyzing either regime, we characterize the fully-informed, fully-aligned optimum:

$$(m^{FB}(\kappa), a^{FB}(\kappa)) = \arg \max_{m,a} U(m, a; \kappa).$$

**Proposition 3** (First-best is a threshold rule in  $\kappa$ ). *On the interior domain  $\kappa \geq \Lambda/k$ ,  $U$  is linear in  $a$  for fixed  $m$ , so  $a^{FB}(\kappa) \in \{0, 1\}$ . Define  $\kappa^* = (\Lambda + c_Q)/(2k)$ . If  $\Lambda > c_Q$ , then  $a^{FB}(\kappa) = 1$  (always ask) for  $\kappa > \kappa^*$  and  $a^{FB}(\kappa) = 0$  (never ask, guess) for  $\kappa < \kappa^*$ .*

*Proof.* At  $a = 1$ :  $m^*(\kappa; 1) = c_Q/\kappa$  and  $U(m^*, 1; \kappa) = V - c_Q k + c_Q^2/(2\kappa)$ . At  $a = 0$ :  $m^*(\kappa; 0) = \Lambda/\kappa$  and  $U(m^*, 0; \kappa) = V - \Lambda k + \Lambda^2/(2\kappa)$ . Subtracting,

$$U(a=1) - U(a=0) = [\Lambda - c_Q] \left( k - \frac{\Lambda + c_Q}{2\kappa} \right),$$

which is positive iff  $\Lambda > c_Q$  and  $\kappa > (\Lambda + c_Q)/(2k) = \kappa^*$ .  $\square$

**Remark 3** (Boundary types  $\kappa < \Lambda/k$ ). *For low-cost types with  $\kappa < \Lambda/k$ , unconstrained first-best effort exceeds  $k$ , so specification saturates at  $m^{FB} = k$ . At  $m = k$ , all attributes are resolved upstream ( $k - m = 0$ ), rendering the asking policy payoff-irrelevant ( $U(k, a; \kappa) = V - \kappa k^2/2$ ).*

This is a clean, closed-form, type-dependent first best: high-cost (terse) users, who cannot cheaply self-provide detail, should always be asked; low-cost (verbose) users, who can cheaply write enough that the small residual is safe to guess, should never be asked. This threshold rule completely characterizes the efficient asking policy as a function of the user's marginal cost of detail relative to downstream error loss.

## 5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate  $a \in [0, 1]$  and cannot condition on user-specific attributes or observed effort. This provides a natural benchmark for policies embedded uniformly at training time or applied without user-level profiling.

### 5.1 User best response

The FOC  $\partial U / \partial m = \Lambda(1 - a) - \kappa m + a c_Q = 0$  gives, with  $\beta(\kappa) = \Lambda / \kappa$  and  $\gamma(\kappa) = (\Lambda - c_Q) / \kappa$ ,

$$m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa), \quad \frac{\partial m^*}{\partial a} = -\gamma(\kappa) = \frac{c_Q - \Lambda}{\kappa}.$$

We assume interior specification levels  $m^* \in (0, k)$ ; for low-cost types where unconstrained  $\beta(\kappa) \geq k$ , specification effort saturates at  $m^* = k$ . The closed-form population moments  $\bar{R}_0$  and  $\bar{\gamma}$  below are exact when best responses remain interior, and serve as tractable linear approximations when boundary clipping binds; as verified in Section 10 and our computational robustness suite, the qualitative regime structure and corner-solution properties are strictly preserved under boundary clipping and exact payoffs alike.

**Remark 4** (Non-monotone effect of ask rate on specification). *Unlike Corollary 1, the sign of  $\partial m^* / \partial a$  here does not depend on type: it is positive for every type iff  $c_Q > \Lambda$ , i.e., iff answering a question costs users more than the expected loss it avoids. When that condition holds, a higher population ask rate induces users to specify more, not less; they pre-empt costly clarifying questions by articulating the answer directly. This introduces a new economic force relative to Section 2, where  $a$  operated solely through effective accuracy, and generates a distinct, directly testable prediction (prediction P1' below).*

### 5.2 The leader's problem

Let  $R(a, \kappa) = k - m^*(\kappa; a)$  denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate  $a \in [0, 1]$  applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.2 at the follower's best response  $m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa)$ :

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A U(m^*(\kappa; a), a; \kappa) - (\lambda_A - c_Q) a (k - m^*(\kappa; a)).$$

Substituting  $U(m^*(\kappa; a), a; \kappa) = V - \Lambda(1 - a)(k - m^*) - \frac{\kappa}{2}(m^*)^2 - a c_Q (k - m^*)$ , using  $k - m^*(\kappa; a) = R_0(\kappa) + a \gamma(\kappa)$ , and defining the effective resolution stake  $s \equiv \Lambda - c_Q$  and the assistant's excess asking friction  $b \equiv \lambda_A - c_Q$ , expanding in powers of  $a$  yields:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A \left( V - \Lambda k + \frac{\Lambda^2}{2\kappa} \right) + (\mu_A s - b) R_0(\kappa) a + \frac{(\mu_A s - 2b) \gamma(\kappa)}{2} a^2.$$

Averaging over  $\kappa \sim F$  requires no moments beyond the population sufficient statistics already defined,  $\bar{R}_0 = k - \mathbb{E}[\beta]$  and  $\bar{\gamma} = \mathbb{E}[\gamma] = s \mathbb{E}[1/\kappa]$ :

$$\Pi(a) = \text{const} + \bar{L} a + \bar{Q} a^2,$$

where  $\text{const} = \mu_A (V - \Lambda k) + \frac{\mu_A \Lambda^2}{2} \mathbb{E}[1/\kappa]$ , and

$$\bar{L} \equiv (\mu_A s - b) \bar{R}_0, \quad \bar{Q} \equiv \frac{(\mu_A s - 2b) \bar{\gamma}}{2}.$$

**Proposition 4** (Stackelberg-optimal pooling rate). *Provided  $\bar{Q} < 0$  (the second-order condition for strict concavity, equivalent to  $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$ ), the leader's unconstrained stationary point is*

$$a^{SE} = -\frac{\bar{L}}{2\bar{Q}} = -\frac{(\mu_{AS} - b)\bar{R}_0}{(\mu_{AS} - 2b)\bar{\gamma}}.$$

*When this stationary point lies in  $[0, 1]$ , it is the unique global maximum of  $\Pi(a)$  on  $[0, 1]$ .*

**Corollary 2** (Bias shifts the pooling rate on the interior branch). *Differentiating the closed form in Proposition 4 directly with respect to  $\lambda_A$  (holding primitives  $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$  fixed) gives, on the interior branch  $\bar{Q} < 0$ ,*

$$\frac{\partial a^{SE}}{\partial \lambda_A} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_{AS}}{(\mu_{AS} - 2b)^2} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A(\Lambda - c_Q)}{[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)]^2},$$

*which is strictly positive whenever  $\bar{R}_0 < 0$  (the under-specification regime). Hence,  $a^{SE}$  is strictly increasing in perceived friction  $\lambda_A$  throughout the interior branch. This comparative static holds on the interior branch ( $\bar{Q} < 0$ ); for the global optimum in the corner regime, see Corollary 4.*

**Corollary 3** (Non-identification of the two bias channels in the interior regime). *On the interior branch ( $\bar{Q} < 0$ ), defining the normalized excess friction per unit of altruism  $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A} = \frac{b}{\mu_A}$ , the optimal pooling rate simplifies to*

$$a^{SE} = -\frac{\bar{R}_0}{\bar{\gamma}} \left[ \frac{s - \tilde{\rho}}{s - 2\tilde{\rho}} \right],$$

*so  $a^{SE}$  depends on the two bias parameters  $(\mu_A, \lambda_A)$  strictly through the single scalar ratio  $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$  (holding primitives  $s$  and  $\bar{R}_0/\bar{\gamma}$  fixed). Consequently,*

$$\frac{\partial a^{SE}/\partial \mu_A}{\partial a^{SE}/\partial \lambda_A} = -\frac{\lambda_A - c_Q}{\mu_A} = -\tilde{\rho},$$

*and any two parameter pairs on the same ray  $(\lambda_A - c_Q)/\mu_A = \text{const}$  produce the identical pooling ask rate. A leader that discounts downstream welfare ( $\mu_A < 1$ ,  $\lambda_A = c_Q \implies \tilde{\rho} = 0$ ) cannot be distinguished, from  $a^{SE}$  alone, from any leader along the corresponding ray of excess friction.*

**Remark 5** (Scope of Corollary 3). *The derivation above is stated on the interior branch  $\bar{Q} < 0$ . By Corollary 5 below, the unbiased pooling optimum is always a corner ( $\bar{Q} \geq 0$ ), so the interior stationary point describes behavior when friction bias is sufficiently large ( $2b > \mu_{AS}$ ).*

**Corollary 4** (Ray-invariance extends to the corner regime). *On the corner branch ( $\bar{Q} \geq 0$ , or whenever the unconstrained stationary point lies outside  $[0, 1]$ ), the optimal policy is chosen by the sign of the corner difference  $\Pi(1) - \Pi(0) = \bar{L} + \bar{Q}$ :*

$$\Pi(1) - \Pi(0) = \mu_A \left[ s \left( \bar{R}_0 + \frac{\bar{\gamma}}{2} \right) - \tilde{\rho}(\bar{R}_0 + \bar{\gamma}) \right].$$

*Since  $\mu_A > 0$ ,  $\mu_A$  factors out completely, and the sign depends solely on the scalar ratio  $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$ . The critical threshold ratio is*

$$\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}},$$

with  $a^{SE} = 1$  for  $\tilde{\rho} < \tilde{\rho}^*$  and  $a^{SE} = 0$  for  $\tilde{\rho} > \tilde{\rho}^*$ . Ray-invariance in the excess friction ratio  $\tilde{\rho}$  therefore holds unconditionally across the entire parameter space, interior and corner regimes alike. A numerical sweep of 11 rays across 10 values of  $\mu_A \in [0.1, 1.0]$  (110 evaluations) confirms this analytical threshold with zero boundary-crossing violations.

**Corollary 5** (Pooling is a corner solution in the unbiased case). *In the unbiased case ( $\mu_A = 1, \lambda_A = c_Q \implies b = 0$ ), the quadratic coefficient is:*

$$\bar{Q} = \frac{s^2 \bar{\gamma}}{2} = \frac{(\Lambda - c_Q)^2 \mathbb{E}[1/\kappa]}{2} \geq 0 \quad \text{always.}$$

The second-order condition for an interior maximum therefore fails:  $\Pi(a)$  is weakly convex on  $[0, 1]$ , its stationary point is a local minimum, and the true optimum is always a corner,  $a^{SE} \in \{0, 1\}$ , not an interior compromise. Comparing the two corners directly:

$$\Pi(1) - \Pi(0) = \bar{L} + \bar{Q} = s \left( \bar{R}_0 + \frac{\bar{\gamma}}{2} \right) = (\Lambda - c_Q) \left[ k - \frac{\Lambda + c_Q}{2} \mathbb{E} \left[ \frac{1}{\kappa} \right] \right],$$

so the unbiased pooling optimum is  $a^{SE} = 1$  (always ask) when  $k \geq \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$  and  $a^{SE} = 0$  (never ask) when  $k < \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$ . This threshold matches the population-averaged first-best preference  $\mathbb{E}[U(1) - U(0)]$  from Proposition 3 identically.

This property proves essential for the comparative analysis in Section 7: Model I exhibits two independent, additive sources of welfare loss relative to the first best: the structural pooling restriction and objective bias.

## 6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types  $\kappa_L < \kappa_H$  with population shares  $f_L, f_H$  (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles  $(m_L, a_L), (m_H, a_H)$  subject to

$$\text{IC}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \quad \text{IR}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize  $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$ .

### 6.1 Single-crossing

The marginal rate of substitution between  $m$  and  $a$  along an indifference curve is  $\text{MRS}(m, a; \kappa) = -(\partial U / \partial m) / (\partial U / \partial a)$ . From user utility  $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2} m^2 - ac_Q(k - m)$ , we have  $\partial U / \partial m = \Lambda(1 - a) + ac_Q - \kappa m$  and  $\partial U / \partial a = s(k - m)$ , where  $s \equiv \Lambda - c_Q > 0$  is the net social gain from asking. Hence,

$$\text{MRS}(m, a; \kappa) = \frac{\kappa m - [\Lambda - as]}{s(k - m)}, \quad \frac{\partial \text{MRS}}{\partial \kappa} = \frac{m}{s(k - m)} > 0,$$

which holds strictly for all interior specification levels  $0 < m < k$  under the maintained condition  $s > 0$ . Higher- $\kappa$  types therefore have strictly steeper indifference curves in  $(m, a)$ -space: they require more compensating ask-service per unit of specification effort given up. This satisfies the standard Spence–Mirrlees single-crossing condition (Spence, 1973) and underwrites the screening analysis below.

## 6.2 No distortion without bias

**Proposition 5** (Efficiency survives private information when the leader is unbiased). *If  $\mu_A = 1$  and  $\lambda_A = c_Q$  (so  $\Pi_\kappa \equiv U(\cdot; \kappa)$  for every  $\kappa$ ), then implementing each type's own first-best bundle,  $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$  from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

*Proof.*  $(m^{FB}(\kappa), a^{FB}(\kappa))$  maximizes  $U(\cdot; \kappa)$  by definition, so for any  $\kappa' \neq \kappa$ ,  $U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \geq U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$  trivially: no bundle, including another type's, can give type  $\kappa$  more utility than its own maximizer. IC holds with no distortion needed. Since  $\Pi_\kappa \equiv U(\cdot; \kappa)$ , the leader's payoff from serving each true type is exactly maximized at that type's own first best, so this menu is also leader-optimal.  $\square$

This result highlights a fundamental distinction between human–AI coordination and classical adverse selection (such as Baron–Myerson or Laffont–Tirole). In regulatory and auction models, the principal pays monetary transfers to extract surplus and thus treats informational rents as a pure loss. Here, by contrast, the leader's objective can coincide directly with user welfare; when it does, incentive compatibility is costless, and private information incurs no welfare penalty. Distortion in this setting is therefore a pure signature of objective misalignment, not of asymmetric information per se.

## 6.3 Distortion under bias

When  $\Pi_\kappa \neq U(\cdot; \kappa)$  (bias present), the leader's preferred bundle per type,

$$(m_\kappa^B, a_\kappa^B) = \arg \max_{m,a} \Pi_\kappa(m, a)$$

(computable in closed form by the same method as Proposition 3, with  $\Lambda, c_Q$  replaced by their  $\mu_A, \lambda_A$ -weighted counterparts), no longer automatically satisfies IC, because  $\Pi_\kappa$  and the user's true  $U(\cdot; \kappa)$  now rank bundles differently.

**Proposition 6** (Downward distortion of the high-cost type's service). *Assume  $\Lambda > c_Q$ ,  $\kappa_L < \kappa^{*B} < \kappa_H$ , and under-asking bias ( $\lambda_A > \mu_A c_Q$ , so  $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A > c_Q$ ). Then:*

1. **Incentive violation of unconstrained menu:** *The unconstrained biased menu  $(m_L^B, a_L^B) = (\Lambda/\kappa_L, 0)$  and  $(m_H^B, a_H^B) = (c_Q^{eff}/\kappa_H, 1)$  strictly violates  $IC_L$  whenever the type heterogeneity spread  $\kappa_H - \kappa_L$  is sufficiently large.*
2. **Downward distortion:** *Under the optimal incentive-compatible menu, the low-cost type receives no distortion ( $a_L^{SB} = a_L^B = 0$ ), while the high-cost type's ask rate is strictly distorted downward:*

$$a_H^{SB} < a_H^B \leq a_H^{FB} = 1.$$

3. **Active constraint regime:** *In the moderate-bias regime,  $IC_L$  binds alone while  $IR_H$ ,  $IC_H$ , and  $IR_L$  remain strictly slack. Under sufficiently extreme bias,  $IC_H$  can become co-active with  $IC_L$ , and the optimal menu may collapse to a pooling or boundary-saturated allocation.*

*Proof.* We establish the result in five analytical steps:

**Step 1: Unconstrained biased optima.** The leader objective per type is  $\Pi_\kappa(m, a) = \mu_A[V - \Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k-m)]$ , where  $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A > c_Q$  under under-asking bias. Maximizing unconstrained with respect to  $m$  yields  $m^B(\kappa; a) = [\Lambda(1-a) + ac_Q^{eff}]/\kappa$ . Because  $\Pi_\kappa$  is linear in  $a$  for fixed  $m$  with slope  $\mu_A(\Lambda - c_Q^{eff})(k-m)$ , the biased first-best ask rate is bang-bang around the threshold  $\kappa^{*B} = (\Lambda + c_Q^{eff})/(2k)$ . For  $\kappa_L < \kappa^{*B} < \kappa_H$ , the unconstrained menu is  $(m_L^B, a_L^B) = (\Lambda/\kappa_L, 0)$  and  $(m_H^B, a_H^B) = (c_Q^{eff}/\kappa_H, 1)$ .

**Step 2: Upward temptation to mimic (IC<sub>L</sub> violation).** Evaluating type  $\kappa_L$ 's payoff from accepting the high type's bundle  $(m_H^B, 1)$  versus its own bundle  $(m_L^B, 0)$ :

$$U(m_H^B, 1; \kappa_L) - U(m_L^B, 0; \kappa_L) = \frac{\kappa_H - \kappa_L}{2}(m_H^B)^2 > 0.$$

Because type  $\kappa_L$  can provide the high type's specification level  $m_H^B = c_Q^{eff}/\kappa_H$  at strictly lower cognitive cost ( $\kappa_L < \kappa_H$ ), its utility from mimicking type  $H$  exceeds type  $H$ 's own utility by the informational rent  $\frac{\kappa_H - \kappa_L}{2}(m_H^B)^2$ . When  $\kappa_H - \kappa_L$  is sufficiently wide,  $U(m_H^B, 1; \kappa_L) > U(m_L^B, 0; \kappa_L)$ , violating IC<sub>L</sub>.

**Step 3: Slackness of participation (IR<sub>H</sub>).** In classical adverse selection with monetary transfers (Baron–Myerson, Laffont–Tirole), the principal's payoff is linearly reduced by agent rent, driving the inefficient agent's IR to bind at reservation  $\underline{U}$ . Here, the principal's payoff  $\Pi_{\kappa_H}$  directly incorporates  $\mu_A U(\cdot; \kappa_H)$ . Depressing  $U_H$  directly penalizes the leader's own objective. Thus, the leader adjusts the bundle  $(m_H, a_H)$  only as far as necessary to restore IC<sub>L</sub>, leaving IR<sub>H</sub> strictly slack:  $U(m_H^{SB}, a_H^{SB}; \kappa_H) > \underline{U}$ .

**Step 4: Global optimization along the IC<sub>L</sub> boundary.** When IC<sub>L</sub> binds, any feasible allocation must satisfy  $U(m_H, a_H; \kappa_L) \leq U(m_L, a_L; \kappa_L)$ . First, consider the low-cost bundle  $(m_L, a_L)$ . Because  $(m_L^B, a_L^B) = (\Lambda/\kappa_L, 0)$  is the unique unconstrained maximizer of both  $U(\cdot; \kappa_L)$  and  $\Pi_{\kappa_L}$  on the relevant domain, any distortion of  $(m_L, a_L)$  away from  $(m_L^B, 0)$  would strictly reduce  $U(m_L, a_L; \kappa_L)$ , tightening IC<sub>L</sub> and reducing leader surplus. Hence, standard no distortion at the top applies to the low-cost type:  $(m_L^{SB}, a_L^{SB}) = (m_L^B, 0)$ , fixing type  $L$ 's utility at  $U_L^0 \equiv U(m_L^B, 0; \kappa_L) = V - \Lambda k + \frac{\Lambda^2}{2\kappa_L}$ .

The entire burden of satisfying IC<sub>L</sub> must therefore fall on the high-cost bundle  $(m_H, a_H)$ . Along the binding boundary  $U(m_H, a_H; \kappa_L) = U_L^0$ , utility is strictly linear in  $a_H$ :

$$V - \Lambda(k - m_H) + a_H s(k - m_H) - \frac{\kappa_L}{2}m_H^2 = U_L^0,$$

where  $s \equiv \Lambda - c_Q > 0$ . Because  $s(k - m_H) > 0$  for all  $m_H < k$ , this uniquely determines the required asking rate  $a_H$  as a function of specification effort  $m_H$ :

$$a_H(m_H)(k - m_H) = \frac{1}{s} \left[ U_L^0 - V + \Lambda(k - m_H) + \frac{\kappa_L}{2}m_H^2 \right],$$

or equivalently,

$$a_H(m_H) = 1 - \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)}.$$

Next, evaluate the leader's payoff  $\Pi_{\kappa_H}(m_H, a_H) = \mu_A U(m_H, a_H; \kappa_H) - b a_H(k - m_H)$  along this boundary, where  $b \equiv \lambda_A - c_Q > 0$ . Using the rent identity  $U(m_H, a_H; \kappa_H) = U_L^0 - \frac{\kappa_H - \kappa_L}{2}m_H^2$  and

substituting  $a_H(m_H)(k - m_H)$ , the leader's reduced objective becomes an exact function of  $m_H$  alone:

$$\Pi_{\kappa_H}(m_H) = \mu_A \left[ U_L^0 - \frac{\kappa_H - \kappa_L}{2} m_H^2 \right] - \frac{b}{s} \left[ U_L^0 - V + \Lambda(k - m_H) + \frac{\kappa_L}{2} m_H^2 \right].$$

Expanding terms:

$$\Pi_{\kappa_H}(m_H) = \left( \mu_A - \frac{b}{s} \right) U_L^0 + \frac{b}{s} (V - \Lambda k) + \frac{b\Lambda}{s} m_H - \frac{1}{2} \left[ \mu_A(\kappa_H - \kappa_L) + \frac{b}{s} \kappa_L \right] m_H^2.$$

The second derivative with respect to  $m_H$  is:

$$\frac{d^2 \Pi_{\kappa_H}}{dm_H^2} = - \left[ \mu_A(\kappa_H - \kappa_L) + \frac{b}{s} \kappa_L \right] = - \frac{\mu_A s (\kappa_H - \kappa_L) + b \kappa_L}{s} < 0,$$

which is strictly negative everywhere on  $[0, k]$  because  $\kappa_H > \kappa_L$ ,  $\mu_A > 0$ ,  $s > 0$ , and  $b > 0$ . Therefore, along the binding  $IC_L$  boundary, the leader's objective is globally strictly concave in  $m_H$ , with unique interior maximizer:

$$m_H^* = \frac{b\Lambda}{\mu_A s (\kappa_H - \kappa_L) + b \kappa_L} = \frac{\Lambda \tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho} \kappa_L},$$

where  $\tilde{\rho} \equiv b/\mu_A = (\lambda_A - c_Q)/\mu_A$ .

Evaluating the asking rate at this global boundary optimum:

$$a_H^{SB} = a_H(m_H^*) = 1 - \frac{U(m_H^*, 1; \kappa_L) - U_L^0}{s(k - m_H^*)}.$$

For sufficiently heterogeneous types,  $U(m_H^*, 1; \kappa_L) > U_L^0$ , which strictly forces:

$$a_H^{SB} < a_H^B \leq a_H^{FB} = 1.$$

In our representative calibration ( $k = 10, \Lambda = 5, c_Q = 2, s = 3, b = 2, \mu_A = 1, \kappa_L = 0.3, \kappa_H = 0.5$ ), this yields  $m_H^* = 25/3 \approx 8.33$  and  $a_H^{SB} = 0.75 < 1$ , exactly matching the numerical second-best contract.

Finally, maintaining  $a_H = 1$  is globally strictly suboptimal: along the boundary  $a_H = 1$ , satisfying  $IC_L$  would require choosing an extreme specification level  $m_H$  such that  $U(m_H, 1; \kappa_L) \leq U_L^0$ , which departs substantially from  $m_H^B$ , destroys cognitive surplus, and incurs full asking friction  $b(k - m_H)$ , yielding strictly lower leader surplus than  $(m_H^*, a_H(m_H^*))$ . Hence, the high-cost type's asking service is strictly distorted downward:  $a_H^{SB} < a_H^B \leq 1$ .

**Step 5: Co-activity of  $IC_H$  under extreme bias.** As bias  $\lambda_A$  increases, maintaining  $IC_L$  forces  $a_H^{SB}$  progressively lower. When  $a_H^{SB}$  approaches  $a_L^{SB} = 0$ , the high-cost type receives negligible clarification assistance relative to its higher specification cost. Eventually, the reverse incentive constraint  $IC_H$  ( $U(m_H, a_H; \kappa_H) \geq U(m_L, a_L; \kappa_H)$ ) can become co-active. Under sufficiently extreme bias, separating the types via distinct asking rates is no longer incentive-viable, and the optimal screening menu may collapse to a pooling or boundary-saturated allocation where both types receive identical asking service.  $\square$

**Remark 6** (Active constraint structure without monetary transfers). *In classical transfer-based screening (e.g., Baron–Myerson),  $IC_L$  and  $IR_H$  typically bind while  $IC_H$  and  $IR_L$  remain slack. In our setting, solving the full four-variable constrained problem shows that  $IC_L$  binds alone, with  $IR_H$ ,  $IC_H$ , and  $IR_L$  remaining strictly slack at representative under-asking-bias parameters. This structural divergence arises because the leader’s payoff  $\Pi_{\kappa_H}$  directly internalizes follower welfare  $\mu_A U(\cdot; \kappa_H)$ , eliminating the classical principal’s motive to compress agent rent down to the reservation utility. We note that the exact active constraint configuration is parameter-region-specific: under sufficiently extreme asking friction ( $\lambda_A \gg c_Q$ ) or severe discounting ( $\mu_A \ll 1$ ),  $IC_H$  can bind alongside  $IC_L$ . The robust qualitative insight is that screening under misalignment strictly distorts  $a_H^{SB} < a_H^B$ , while the active constraint configuration reflects the non-monetary, partially aligned nature of the objective.*

**Remark 7.** *The direction in Proposition 6 is specific to the under-asking bias case ( $\lambda_A > \mu_{ACQ}$ ), which we take to be the empirically relevant one (Section 3.3’s motivating story: turn-level satisfaction signals penalize extra clarifying questions more than users actually mind answering them). Under the opposite bias, the direction of distortion can flip; we do not claim a sign that holds for all  $\Pi_{\kappa}$ .*

## 7 Comparing the Two Regimes

**Corollary 6** (Menus weakly dominate pooling). *The leader’s optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy  $a$  is a feasible (degenerate) menu with  $a_L = a_H = a$ , trivially satisfying  $IC$ , so the unconstrained menu optimum is at least as good.*

Beyond payoff dominance, the central comparison lies in the structural decomposition of welfare losses across the two regimes, enabled by Propositions 4–6:

- **Model I (pooling)** incurs welfare loss from two distinct sources: (i) the pooling restriction itself, which operates even when  $\mu_A=1$  and  $\lambda_A=c_Q$ , where it manifests as a corner solution that fully serves one type while completely neglecting the other (Corollary 5); and (ii) objective bias, which shifts  $a^{SE}$  (Corollary 2) and determines whether the equilibrium settles at a corner or an interior rate. These two forces interact non-linearly across the parameter space.
- **Model II (screening)** incurs welfare loss from bias *alone* (Proposition 5): in the absence of bias, screening recovers full efficiency despite private cost information. Under bias, the loss takes the form of the downward distortion characterized in Proposition 6. This represents a fundamentally different mechanism from the pooling loss: it stems not from an inability to condition on observed effort, but from the requirement to prevent opportunistic mimicking across types.

### 7.1 Sensitivity of the Unbiased Pooling Corner

While the qualitative finding of Corollary 5, that unbiased pooling is always a corner solution rather than an interior compromise, holds universally across the parameter space, the specific corner selected depends on whether task complexity  $k$  exceeds the population threshold  $k^* \equiv \frac{\Lambda+c_Q}{2} \mathbb{E}[1/\kappa]$ . Table 1 illustrates this boundary across varying task sizes  $k$  and baseline guessing accuracies  $g$ , holding  $L = 10$ ,  $c_Q = 2$ , and  $\kappa \sim \text{Uniform}[0.2, 0.6]$  (where  $\mathbb{E}[1/\kappa] \approx 2.747$ ).

**Remark 8** (Compounding, testable). *Because Model II’s distortion is triggered by, and scales with, the same bias parameters ( $\mu_A, \lambda_A$  vs.  $c_Q$ ) that drive Model I’s shift in  $a^{SE}$ , reducing measured*

| $g$  | $\Lambda = L(1 - g)$ | Threshold $k^*$ | Task Size $k$  | $\Pi(1) - \Pi(0)$ | Optimal Corner $a^{SE}$ |
|------|----------------------|-----------------|----------------|-------------------|-------------------------|
| 0.30 | 7.0                  | 12.36           | 5              | -36.80            | 0 (never ask)           |
| 0.30 | 7.0                  | 12.36           | 10             | -11.80            | 0 (never ask)           |
| 0.30 | 7.0                  | 12.36           | 15             | +13.20            | 1 (always ask)          |
| 0.50 | 5.0                  | 9.61            | 5              | -13.84            | 0 (never ask)           |
| 0.50 | 5.0                  | 9.61            | 9              | -1.84             | 0 (never ask)           |
| 0.50 | 5.0                  | 9.61            | 10 (Benchmark) | +1.16             | 1 (always ask)          |
| 0.50 | 5.0                  | 9.61            | 12             | +7.16             | 1 (always ask)          |
| 0.50 | 5.0                  | 9.61            | 15             | +16.16            | 1 (always ask)          |
| 0.70 | 3.0                  | 6.87            | 5              | -1.87             | 0 (never ask)           |
| 0.70 | 3.0                  | 6.87            | 8              | +1.13             | 1 (always ask)          |
| 0.70 | 3.0                  | 6.87            | 10             | +3.13             | 1 (always ask)          |

Table 1: Sensitivity of the unbiased pooling corner choice  $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$  across task complexity  $k$  and guessing capability  $g$ , for  $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$ . At the paper’s canonical benchmark ( $g = 0.5, k = 10$ ), asking yields a modest advantage of +1.16, which flips to never-asking at  $k \leq 9$ .

*assistant bias should improve outcomes for high-cost (terse) users under both regimes, but the welfare gap between menu-conditioned and pooled policies should track population heterogeneity in  $\kappa$  (the spread  $\kappa_H - \kappa_L$  and how far  $F$  sits on either side of  $\kappa^*$ ) almost independently of bias. This gives two separable, testable quantities from the same field data: a bias-sensitive term and a heterogeneity-sensitive term.*

## 8 Testable Predictions

- P1** (Prop. 1, Cor. 1) Holding the assistant’s policy fixed, higher guessing accuracy is associated with shorter specifications alongside higher or equal success rates, and never with shorter specifications paired with lower success rates.
- P1’** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length flips sign depending on whether  $c_Q \gtrless \Lambda = L(1 - g)$ : where clarifying questions are inexpensive relative to the risk they resolve, users specify less as the ask rate rises (substitution); where questions are relatively costly to answer, users specify *more* (pre-emption). This prediction operates within the model and is distinct from P1’s accuracy channel.
- P2** (Prop. 2) The correlation between turn-level satisfaction and eventual task success weakens or inverts specifically for under-asking assistants, and can be benchmarked against the derived  $a^{SE}$  of Proposition 4 rather than an assumed ask rate.
- P3** Making ask-rate statistics visible to users shifts specification length toward the sophisticated best response analyzed in Section 3.3.
- P4** (Prop. 3) Ask rates should rise with a user’s revealed cost of providing detail (terseness), exhibiting a sharp transition around the threshold  $\kappa^*$ , rather than responding exclusively to model confidence.
- P5** Users with lower self-reported error-catching rates exhibit declining specification effort across an interactive session (following Section 3’s dynamic channel).

- P6** (Cor. 3) Under a pooling policy, terse users are under-served and verbose users are over-served relative to *each type’s own* first best, even for a fully unbiased assistant. This efficiency gap is attributable purely to instrument restriction; it can be distinguished empirically from bias by holding  $\lambda_A$  fixed (for instance, via controlled variation in training friction penalties) and varying only whether the policy can condition on declared specification effort.
- P7** (Prop. 6) Even when an assistant *can* condition on declared specification effort, terse users receive fewer clarifying questions than the assistant’s own per-type preference would prescribe. This distortion arises specifically from the incentive-compatibility constraint and contracts as population cost heterogeneity ( $\kappa_H - \kappa_L$ ) diminishes, holding bias fixed.

## 9 Empirical Strategy and Hierarchy of Evidence

To maintain clear scientific traceability, the theoretical claims and computational evaluations in this paper are organized into five distinct levels of evidence:

1. **Level 1: Analytical Theorems.** Core equilibrium characterizations (the First-Best threshold in Proposition 3, the pooling closed form in Proposition 4, unbiased corner pooling in Corollary 5, single-crossing in Section 6, full efficiency in Proposition 5, and downward screening distortion in Proposition 6) proven formally from primitives under the linear-risk specification.
2. **Level 2: Exact Nonlinear Robustness.** Constrained numerical optimizations and grid sweeps demonstrating that the qualitative equilibrium structure (corner pooling, downward distortion, and extreme-bias regimes) survives when replacing the linear-risk approximation with the exact conjunctive success function  $q(a, g)^{k-m}$  (Section 10).
3. **Level 3: Numerical Primitive Verification.** Computational checks in the open-source suite that verify analytical closed forms against fine-grid argmax searches, check second-order conditions, match finite-difference comparative statics to analytical derivatives within 1.8%, and audit active Kuhn–Tucker constraint sets directly from numerical Lagrange multipliers.
4. **Level 4: Falsifiable Behavioral Predictions.** Seven distinct, directional empirical predictions (P1–P7 in Section 8) that translate theoretical mechanisms into observable margins for experimental testing.
5. **Level 5: Computational Simulation Infrastructure.** A software pipeline in the accompanying codebase that instantiates simulated user and assistant interaction protocols under the derived policies, with deterministic stubs for reproducible execution. This layer provides computational verification of the interaction mechanics rather than empirical evidence of human behavior.

**Proposed human-subject study design.** Testing predictions P1’, P6, and P7 requires moving beyond observational log analysis where assistant bias  $\lambda_A$  and user cost  $\kappa$  are confounded. We propose a between-subjects randomized controlled trial on a population of software developers performing standardized coding tasks:

- *Treatments:* Users are randomly assigned to assistants operating under: (A) Low asking policy ( $a = 0.1$ ), (B) High asking policy ( $a = 0.8$ ), (C) Menu-conditioned screening policy (where prompt length endogenously selects the clarification rate), and (D) Unbiased benchmark.

- *Measurements:* For each task, we log upstream prompt length, number of specified edge cases/invariants, turn-level clarification counts, downstream test suite pass rate, and post-session subjective satisfaction.
- *Identification:* By exogenously varying  $a$  and measuring whether terse users expand prompt effort when questions are expensive ( $c_Q > \Lambda$ , testing P1'), this design isolates the causal effect of clarification policies on human specification economizing.

## 10 Discussion and Limitations

The linear-risk approximation to the conjunctive success function is the main tractability assumption carried through Sections 4–7. To verify that this simplification does not drive our core theoretical findings, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff  $q(a, g)^{k-m}$  (independently of the linear-risk closed forms, via direct numerical grid search and constrained optimization). Across our tested parameter grid ( $k \in [3, 20]$  and  $g \in [0.50, 0.95]$ ), numerical optimization confirms that the qualitative results survive under the exact nonlinear specification: the corner-solution property of Corollary 5 survives in 100% of tested configurations ( $a_{\text{exact}}^{SE} \in \{0, 1\}$ ); downward distortion ( $a_H^{SB} < a_H^B$ ) persists across the grid, with exceptions concentrated at small  $k$  where specification effort saturates at  $m = k$ ; and participation constraints remain strictly slack. Furthermore, a parameter sweep of the full four-variable constrained menu problem under the exact conjunctive payoff across  $\lambda_A \in [c_Q, 10c_Q]$  and  $\mu_A \in [0.1, 1.0]$  (100 grid points) confirms that  $IC_H$  binds alongside  $IC_L$  under extreme bias. This reproduces the exact three-regime structure (no binding constraint,  $IC_L$  alone, and  $IC_L + IC_H$  together) identified under the linear-risk approximation, with only the boundary between moderate- and extreme-bias regimes shifting slightly (from  $\lambda_A \approx 6.0$  to  $\lambda_A \approx 8.0$  at  $\mu_A = 1.0$ ). The extreme-bias reversal is therefore an intrinsic structural feature of the screening contract space, rather than an artifact of the linear-risk approximation.

Regarding the regularity condition (Assumption 1), computational evaluation confirms that monotonicity of prompt specification ( $m^*(g)$  weakly decreasing in  $g$ ) holds across 100% of tested intervals satisfying  $(k - m) \ln q \geq -1$ . While monotonicity breaks in 96.7% of intervals where the cross-partial turns positive, the survival of monotonicity outside strict submodularity boundaries in lower-stakes regions emerges as a numerical robustness property: Topkis submodularity is a sufficient condition, but empirical monotonicity can survive locally over wider parameter domains.

A stylized modeling assumption is attribute symmetry ( $k$  binary attributes with identical loss  $L$  and guessing probability  $g$ ). In practical software engineering, attributes exhibit substantial loss heterogeneity: omitting an architectural invariant or security convention causes severe failure ( $L_i \gg 0$ ), while omitting a stylistic naming convention carries negligible cost ( $L_j \approx 0$ ). In an extension with attribute-specific losses  $L_i$ , the user's objective becomes  $U = V \prod_i q_i - c(m) - \sum_i a_i c_Q$ . Ordering attributes by descending risk stake  $\Lambda_i = L_i(1 - g_i)$ , users rationally specify the highest-stakes attributes first, while assistants prioritize asking on the highest-residual-stake attributes. The fundamental contract-theoretic mechanism, namely the substitution between upstream specification effort and automated clarification under misaligned asking friction, generalizes directly to the heterogeneous-loss setting.

A related limitation concerns the empirical identification of  $\lambda_A$ . Every bias-driven prediction (Corollary 2, Proposition 6, and predictions P2, P6, P7) is stated in terms of  $\lambda_A > \mu_{ACQ}$ , but backing out  $\lambda_A$  from observed policy alone is circular. Corollary 3 formalizes this as an exact non-identification result on the interior branch:  $a^{SE}$  depends on  $(\mu_A, \lambda_A)$  only through their ratio, so a discounting-driven leader and a friction-driven leader with the identical ratio  $\lambda_A/\mu_A$  cannot be

distinguished from pooling behavior alone. An independent empirical handle, such as an explicit penalty on turn count in the training objective or reward specification, would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested without circularity. The two-type restriction in Model II avoids ironing complications at the cost of generality; a continuum formulation using the standard Mirrlees technique is a natural next step that would allow  $\kappa^*$  from the first-best benchmark to interact directly with the screening distortion. We also assume truthful revelation of specified attributes (abstracting from cheap-talk ambiguity within what is communicated) and a single leader. Modeling competition among multiple assistants represents an appealing extension, as market competition could plausibly discipline asking bias directly.

A broader methodological conclusion concerns the limitations of applying classical contract-theoretic intuition directly to human–AI settings without accounting for non-monetary, partially aligned incentives. In classical mechanism design (such as Baron–Myerson or Laffont–Tirole), the principal pays monetary transfers to extract surplus, and agent informational rent directly reduces the principal’s payoff. In human–AI interaction, the assistant’s objective directly incorporates user welfare. As a consequence, incentive compatibility is costless in the absence of bias (Proposition 5), participation constraints do not bind at the bottom of the type distribution, and unbiased pooling naturally induces corner solutions rather than interior compromises (Corollary 5). Economic analyses of automated assistance must therefore explicitly incorporate the non-monetary, partially aligned nature of the interaction rather than relying on direct analogies to cash-transfer mechanisms.

## References

- Heinrich von Stackelberg. *Marktform und Gleichgewicht*. Julius Springer, Berlin/Wien, 1934.
- Roger B. Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.
- David P. Baron and Roger B. Myerson. Regulating a monopolist with unknown costs. *Econometrica*, 50(4):911–930, 1982.
- Michael Mussa and Sherwin Rosen. Monopoly and product quality. *Journal of Economic Theory*, 18(2):301–317, 1978.
- Jean-Jacques Laffont and Jean Tirole. Using cost observation to regulate firms. *Journal of Political Economy*, 94(3):614–641, 1986.
- Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50(6):1431–1451, 1982.
- Robert M. Townsend. Optimal contracts and competitive markets with costly state verification. *Journal of Economic Theory*, 21(2):265–293, 1979.
- Sanford J. Grossman. The informational role of warranties and private disclosure about product quality. *Journal of Law and Economics*, 24(3):461–483, 1981.
- Christopher A. Sims. Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690, 2003.
- Michael Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973.

- Donald M. Topkis. Minimizing a submodular function on a lattice. *Operations Research*, 26(2):305–321, 1978.
- Donald M. Topkis. *Supermodularity and Complementarity*. Princeton University Press, Princeton, NJ, 1998.
- Paul Milgrom and Christina Shannon. Monotone comparative statics. *Econometrica*, 62(1):157–180, 1994.
- Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems* (NIPS), volume 29, pages 3909–3917, 2016.
- Sudha Rao and Hal Daumé III. Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics* (ACL), pages 2737–2746, 2018.
- Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. Asking clarifying questions in open-domain information-seeking conversations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (SIGIR), pages 475–484, 2019.
- Hamed Zamani, Susan Dumais, Nick Craswell, Paul Bennett, and Gord Lueck. Generating clarifying questions for information retrieval. In *Proceedings of the Web Conference 2020* (WWW), pages 418–428, 2020.